The Universe as Spacetime Geometry Projection: The Mystery of the Refresh Delay Field Revealed by the Double-Slit Experiment

An Experiment That Has Haunted Humanity for a Century

More than a century ago, physicist Thomas Young performed the experiment that changed everything. He sent a beam of light through two narrow slits, producing a pattern of bright and dark fringes on a screen behind. This phenomenon is called **interference**—when two light waves meet, they reinforce or cancel each other, creating a unique pattern.

Classically, this is not surprising; wave interference was well understood. But a paradox emerged: when the light source is dimmed so much that photons are emitted **one by one**—each photon an indivisible energy packet—the interference pattern still appears.

How can **one single photon** pass through both slits at the same time and interfere with itself?

This is the core puzzle of the double-slit experiment, which has baffled physicists for more than 100 years.

The Copenhagen school claimed: *Do not ask which slit the photon passed through*—the question is meaningless.

The many-worlds interpretation proposed that each photon has copies in separate universes.

Pilot-wave theories argue that an invisible “guide wave” passes through both slits.

Yet all these interpretations share one assumption: photons possess a mysterious “superposition” ability to exist in two places at once.

What I show you today is this:

We may have been asking the wrong question from the very beginning.

A New Fundamental Variable

Before presenting the new interpretation of the double-slit experiment, we introduce a revolutionary physical framework:

TCM-NAN(λ)

The core of this framework is a single relation:

$$\text{Physical time} = \text{baseline refresh period} + \text{refresh delay field } \lambda \tag{1}$$

The baseline refresh period is on the Planck scale (~10⊠⊠^{3} s), the smallest time unit allowed by the universe.

The field **λ** (lambda) is called the **refresh delay field**.

What is the refresh delay field λ?

Imagine the universe as a vast whiteboard covered with grid points. Every grid point constantly "refreshes" its state—much like a TV screen refreshing frames many times per second. Unlike a screen, this whiteboard is **absolutely rigid**: its spatial coordinates are fixed forever; it does **not** bend or expand.

How long does one refresh take?

In ideal vacuum, the cycle equals the baseline.

But if matter is present, the point requires extra time to refresh.

This **extra time** is **λ**.

The relation is simple:

$$\boxed{\text{Larger } \lambda \Rightarrow \text{slower refresh} \Rightarrow \text{slower time flow}} \tag{2}$$

This single variable drives **all physical phenomena in the universe**.

Gravity Is Simply the Gradient of Refresh Delay

Why does a stone fall? Newton said gravity; Einstein called it spacetime curvature.

In the TCM-NAN framework, the answer is simpler:

Objects are pushed toward regions of slower refresh.

Think of a river: water flows downhill because of elevation differences.

In the world of λ, the "elevation" is **refresh speed**.

Regions with larger λ have slower time and lower "temporal elevation".

Objects are pushed by refresh pressure toward higher-λ regions—just as water flows downhill.

In short:

Gravity is not curved spacetime. It is the spatial gradient of the refresh delay field λ.

The Earth orbits the Sun because the Sun's λ is far larger than the surrounding space. The Earth is continuously pushed toward the Sun while maintaining tangential speed, forming an

approximately circular orbit.

A New Interpretation of the Double-Slit Experiment

We now return to the central puzzle.

In the TCM-NAN(λ) framework, we **never need to ask which slit the photon passed through**.

The λ field is **globally defined**. Before a photon even leaves the source, its entire path is already "preprogrammed" by the λ-field distribution of the full apparatus.

The mechanism is as follows:

- The atoms of the barrier material have **very large λ** (dense electron clouds around nuclei cause extremely slow refresh).
- The vacuum slits have λ near zero (near-ideal vacuum).

When a photon leaves the source, its trajectory is **modulated by the λ field**.

It does not "choose" one slit. Instead, the **effective optical path** of the two routes is differentiated by the λ field, producing observable interference.

Definition: Effective Optical Path

As light propagates, its speed at each point is controlled by local λ.

The effective path length is given by an integral over the trajectory.

Larger λ → light propagates with more "difficulty" → longer effective distance.

Let the effective optical path through the left slit be L_1, and through the right slit be L_2.

When the two paths reach the screen, their phase difference is proportional to $L_1 - L_2$.

- Phase difference = integer multiple of 2π: constructive interference → bright fringe
- Phase difference = odd multiple of π: destructive interference → dark fringe

Thus:

**The interference pattern is not mysterious superposition.

It is simply path-integral differences caused by spatial inhomogeneity of the λ field.**

This is the new explanation of the double-slit experiment—**no quantum superposition required**.

The Deep Meaning of the Single-Photon Experiment

You may ask: How can a single photon form an interference pattern? What is it interfering with?

The answer is clear:

Every photon interacts with the λ field.

In TCM-NAN(λ), the photon’s entire journey—from emission to detection—is continuously modulated by the λ distribution along its path.

There is no mysterious “wavefunction collapse”.

There is only **path-integral convergence at each refresh moment**—the mechanism that fixes the photon’s position.

At the end of each refresh cycle, the photon’s “position” is redetermined.

This process is **not random dice-rolling**. It is **determined by the global λ-field distribution**.

When many photons (or one photon repeated many times) accumulate, their screen positions exactly match the path-integral result from λ-field differences.

Therefore, the single-photon experiment does **not** prove “one particle exists in two states”.

It proves:

The λ field is globally defined. Every particle’s trajectory is pre-determined by the λ distribution.

Delayed-Choice Experiment: Time Does Not Flow Backward

In 1980, physicist John Wheeler designed a radical experiment:

After a photon has already passed the double slits, we decide whether to measure which path it took.

Common sense says measurement can only record the past, not change it.

Yet the result was shocking:

The choice of measurement affected whether interference appeared, as if the present altered the past.

This is the famous **delayed-choice paradox**, one of the weirdest results in quantum mechanics.

In TCM-NAN(λ), the paradox has a clean, rational explanation:

The λ field has global coupling.

The wave equation satisfied by λ allows a special propagation mode:

When the effective mass of λ is nonzero, its influence can be nonzero across spacelike separations.

In other words:

A change in λ here and now can instantly affect distant events—even events that appear “already past”.

To be clear:

The delayed-choice experiment does **not** prove time can reverse.

It proves:

The λ field is a globally correlated whole. Past and present are just neighboring nodes in the refresh sequence—not fixed points on a causal chain.

Four Properties of λ Derived Directly from the Double-Slit Experiment

From the double-slit experiment and its variants, we indirectly "observe" four core properties of the λ field:

Property 1: λ is not uniform

Changing the barrier material noticeably shifts the interference fringes.

This proves λ is **not spatially uniform**:

The barrier region has drastically different λ from vacuum, and its gradient produces observable effects.

Property 2: λ is not static

Even with all parameters perfectly controlled, fringe sharpness has a statistical limit.

This shows λ **fluctuates dynamically in time**; it is not a fixed geometric background.

Property 3: λ refresh is discrete

Interference orders must be integers.

This is not a quantum axiom—it is a mathematical consequence of **discrete refresh**.

Each fringe corresponds to one full phase cycle, which comes from cumulative baseline refresh units.

Property 4: λ has global memory

The delayed-choice experiment proves the λ field can transmit "future information" to "past events".

This is **not** faster-than-light communication.

It is direct evidence of **global correlation of the λ field**.

Full Comparison: TCM-NAN(λ) vs Standard Model

Dimension	Standard Model	TCM-NAN(λ) Framework
Nature of time		Cumulative refresh delay

	Independent background parameter	
Nature of space	Bendable, expandable dynamical metric	Absolutely rigid 3D lattice
Double-slit origin	Probability amplitude superposition	Effective path difference from λ gradient
Quantum superposition	Particle in multiple eigenstates	Global path convergence during discrete refresh
Measurement problem	Wavefunction collapse (axiom)	Refresh forced onto unique branch
Gravity	Spacetime curvature	λ gradient (objects pushed to high λ)
Dark energy	Cosmological constant (assumption)	λ creation source term (inevitable time accumulation)
Redshift mechanism	Space expansion stretches wavelength	Light speed increases with cosmic time
Time dilation	General relativistic effect	Temporal variation of λ
Bell inequality	Rules out local hidden variables	Global coupling of λ Green’s function
Delayed choice	Measurement “affects” past	Global memory of λ field
Energy conservation	Universal law	Local approximation; macroscopic creation

Philosophical Meaning of the Deep Divergence

The core difference is **ontological commitment**: what we believe the universe fundamentally is.

The Standard Model:

The universe is made of “particles” and “fields”.

Time is a flowing background.

Space is an active stage.

TCM-NAN(λ):

The universe is made of **refresh delay**.

Time is the accumulation of refresh delay.

Space is a rigid grid where refresh occurs.

Particles are merely **ripples in the λ field**.

Using the double slit as an example:

- Standard Model: Photon passes both slits in superposition.
- TCM-NAN(λ): Photon path is determined by global λ distribution; interference comes from effective optical-path difference.

A New Worldview

If TCM-NAN(λ) is correct:

- Time is not a river, but the beat of refresh.
- Space is not a stage, but a rigid grid.
- Matter is not the protagonist, but ripples of λ.
- Quantum superposition is not a mystery, but geometry of discrete refresh.
- Energy need not be conserved, because time is a creative force.

Final Remarks

The double-slit experiment has haunted physics for over a century.

TCM-NAN(λ) provides a bold and simple answer:

- No mysterious quantum superposition—only global distribution of λ.
- No weird wave–particle duality—only path integrals of discrete refresh.
- No incomprehensible delayed choice—only global coupling of the λ field.

This framework remains to be tested experimentally:

time drift of the fine-structure constant, luminosity-distance differences between gravitational and electromagnetic waves, echo signals from black-hole mergers…

All are falsifiable predictions.

But one lesson is clear:

Sometimes the deepest breakthrough in physics comes not from more precise measurement,

but from asking a completely new question about what we already see.

What really happens to a photon passing through a double slit?

The answer may lie not in the photon itself,

but in the **ocean of refresh delay** it travels through.

Reference Framework

TCM-NAN(λ), Time Creation Cosmology, Refresh Delay Field Theory

Core Relation

$$\text{Physical time} = \text{baseline refresh period} + \text{refresh delay field } \lambda \tag{3}$$